



Oracle 715am-1 - Sql

Data Structures (Avanthi Institute of Engineering & Technology)



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Notes link:

bit.ly/oracledbnotes

Oracle SQL & PLSQL @ 7:15 AM IST by Mr Shiva Chaitanya sir

Day-1 <https://youtu.be/ZS48xauc34o>

Day-2 <https://youtu.be/1uqtxBq0NqA>

Data Store
Database
DBMS
RDBMS
Metadata

RUN or ANALYZE

BANK

Branches
Customers
Transactions
Products
Creditcards

▪
▪

Opens account
Deposits amount
Withdraws amount
Transfers funds

RUN

In 2023 ?
In 2024 ?

ANALYZE

Amazon online shopping

Products
Customers
Sellers
Orders
Wishlist

▪
▪

Places order
Delivers products

RUN

In 2023 ?
In 2024 ?

ANALYZE

GOAL:

Storing business data permanently.

In java:

Variable:

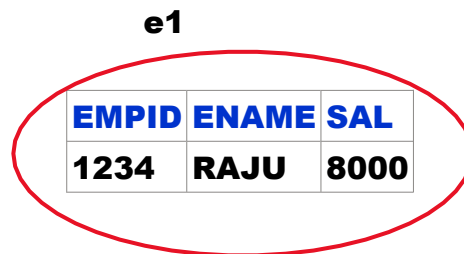
- Variable is used to store the data.
- Variable is temporary.



Object:

- Object is used to store the data.
- Object is temporary.

Employee e1 = new Employee(1234,"RAJU",8000);



File	• It is permanent
Database [best way]	• It is permanent

File	Database
• It is suitable for single user.	• It is suitable for multiple users
• We can store small amounts of data.	• We can store large amounts of data.
• Less security.	• More security.

Data Store:

- Data Store is a location where data is available.

Examples:
BOOKS, FILES, DATABASES

Database:

- Database is a kind of data store.
- Database is a location where organization's business data stored permanently.

Examples:

COLLEGE DB

COURSES
STUDENTS
MARKS
FEE

•
 •

BANK DB

BRANCHES
CUSTOMERS
TRANSACTIONS

•
 •

- Database contains **interrelated** data in an **organized** form.

BANK DB

TABLE Employee

EMPID	ENAME	SAL
1234	KIRAN	15000
1235	SAI	12000

TABLE Customers

Custid	Cname	Ccity	PAN	ACNO

Transactions

TXNI	T_DATE	TIM	T_TYPE	ACN	
D	E			O	

Branches

IFSC_CODE	CITY	STATE	COUNTRY
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DBMS:

- **DBMS => DataBase Management System / Software.**
- **DBMS is a software.**
- **It is used to create and maintain the database.**

Evolution of DBMS:

Before 1960s	Books
In 1960s	FMS => File Management Software
In 1970s	HDBMS => Hierarchical DBMS NDBMS => Network DBMS
In 1976	E.F. Codd => RDBMS Concept
ORACLE company Founder	Larry Ellison
In 1977	Larry Ellison estd a company: Software Development Laboratories
In 1979	Renamed company: Relational Software Inc. Introduced ORACLE s/w
In 1983	Renamed company: ORACLE carp.

RDBMS:

- **RDBMS => Relational DataBase Management System / Software.**
- **RDBMS is a software that is used to create and maintain the database in the form of tables.**

Examples:

ORACLE, SQL SERVER, DB2, Postgre SQL, MY SQL

ORACLE	Is a product of ORACLE company
SQL SERVER	MICROSOFT
DB2	IBM
Postgre SQL	Postgre Forum [a group of companies]
MY SQL	SUN MICRO SYSTEMS [ORACLE]

ORACLE

ORACLE DB SERVER

SQL => QUERIES =>
PL/SQL => PROGRAMS =>

ORACLE

Duration: 2 and half months
Mon to sat
Sundays are holidays

Class duration => 1and half hour

7.20 to 8.50

Google classroom => upload notes

Metadata

ORACLE

SQL
PL/SQL

ORACLE
SQL
PL/SQL

SQL SERVER
SQL
T-SQL

Data Store	Is a location => data Examples: Book, File, database
Database	Data store => loc => org bus data stored permanently
DBMS	Software => to create and maintain the database
RDBMS	DBMS => Software => to create and maintain the database in the form of tables Examples: ORACLE, SQL SERVER, DB2,

TABLE:

- **TABLE** is a collection of rows and columns.

Example:

EMPLOYEE		
EMPID	ENAME	SAL
1234	KIRAN	20000
1235	SAI	15000

Metadata:

- **Metadata** is the data about the data.
- **It tells about the data.**
- **It can be also called as "Data Definition".**

Examples:

Column name, Table name, data type, field size

EMPLOYEE

EMPID Number(4)	ENAME	SAL
1234	KIRAN	20000
RAJU error		
25-DEC-24 error		
123		
123456 ERROR		

ORACLE**Module-1: TABLES [SQL]****Module-2: PL/SQL****Module-3: Other DB Objects [SQL]****Module-1: Tables [SQL]**

SQL commands	DDL, DRL, DML, TCL, DCL
Built-in functions	String functions, Number, date, analytic, aggregate ...
Clauses	Group By, Order By, Having, Offset, Fetch,
Joins	Inner Joins => Equi, Non-equi Outer => left outer, right, full Self cross
Sub queries	Non-correlated single row sq, multi row, inline, scalar Correlated
Constraints	PRIMARY KEY, NOT NULL, UNIQUE, FOREIGN KEY, ...
Set operators	Union, Union all, intersect, minus

Module-2: PL/SQL

PL/SQL Basics	Data types, declare, assign, print, read, Using sql commands in pl/sql
Control Structures	Conditional, Looping, Jumping
CURSORS	Steps, Implicit cursor, Explicit Cursor => Simple cursor, Ref ▪ ▪
COLLECTIONS	Associative array, Nested table, v-array

	Bulk Bind, Bulk Collect
Exception Handling	built-in exceptions, user-defined exceptions Pragma exception_init()
Stored procedures	
Stored functions	
Packages	
Triggers	
Working with LOBS	BFILE, BLOB
Dynamic SQL	

Module-3: Other DB Objects [SQL]

Views	Simple view, Complex view
Indexes	B-tree Index => simple, composite, function-based, unique Bitmap Index
Sequences	Sequence, identity
Materialized views	Refreshing
Synonyms	

SQL DEVELOPER

ORACLE:

- **ORACLE is a Relational DataBase Management Software [RDBMS].**
- **Relation => Table**
- **It is used to create and maintain the database in the form of tables.**
- **Using this, we can store, **manipulate** and **retrieve** the data of database.**

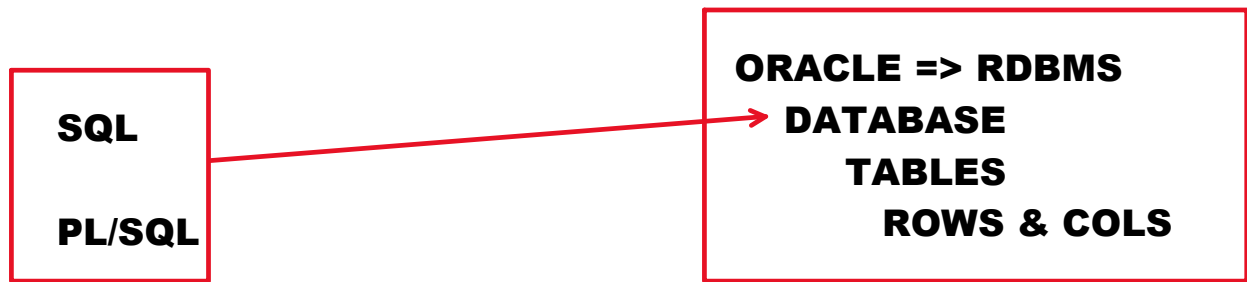
Manipulate => 3 actions => insert / update / delete

Emp joined	=>	INSERT
Emp sal increased	=>	UPDATE
Emp left from org	=>	DELETE

Retrieve => getting back => opening existing data

Checking the balance
Transaction statement
Searching for products

- **In 1979, ORACLE 2nd version introduced. They didn't 1st version.**
- **Latest version is: ORACLE 23ai**



To communicate with ORACLE DB we can use 2 languages. They are:

- **SQL**
- **PL/SQL**

SQL:

- **SQL => Structured Query Language.**
- **SQL is a query language.**
- **In SQL, we develop the queries to communicate with ORACLE DB.**
- **Query => request / command / instruction.**
- **Query is request that is sent to DB SERVER.**
- **SQL is a Non-Procedural Language [no programs].**
It means, in SQL, we will develop any set of statements or programs. Just we develop the queries.
- **SQL is Unified Language.**
It is a common language to work with many relational databases.

ORACLE
DATABASE
↑
TABLES

SQL SERVER
DATABASE
↑
TABLES

DB2
Database
↑
Tables

**DATABASE
TABLES**



SQL

**DATABASE
TABLES**



SQL

**Database
Tables**



SQL

- **SQL provides commands.**
- **SQL provides functions.**

Notes link:

bit.ly/oracledbnotes

BANK DB

Branches Table

IFSC_CODE	CITY	STATE
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Customers

Custid	Cname	Aadhar	PAN	..	..
--------	-------	--------	-----	----	----

Transactions

Txn_Id	T_type	T_date_time	Acno	Amount
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Employees

Empid	Ename	Job	Sal
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-

SQL Commands

Friday, March 7, 2025 7:26 AM



SQL provides 5 sub languages. They are:

- **DDL**
- **DRL / DQL**
- **DML**
- **TCL**
- **DCL / ACL**

DDL • DDL => Data Definition Language • Data Definition => Metadata • It deals with metadata	Create Alter Drop Flashback [oracle 10g] Purge [oracle 10g] Truncate Rename
DRL / DQL • DRL => Data Retrieval Language • DQL => Data Query Language	Select

• Retrieve => opening existing data [getting back] • It deals with data retrievals.	
DML • Data Manipulation Language • Manipulation => 3 actions Insert / update / delete • It deals with data manipulations	Insert Update Delete Insert All [oracle 9i] Merge [oracle 9i]
TCL • Transaction Control Language • It deals with transactions.	Commit Rollback Savepoint
DCL / ACL • Data Control Language • Accessing Control Language • It deals with data accessibility	Grant Revoke

CREATE:

- **CREATE** command is used to create oracle database objects like tables, views, indexes ... etc

ORACLE DB OBJECTS:

TABLES

VIEWS

INDEXES

SEQUENCES

MATERIALIZED VIEWS

SYNONYMS

PROCEDURES

FUNCTIONS

PACKAGES

TRIGGERS

Syntax to create the Table:

```
CREATE TABLE <table_name>  
(  
  <column_name> <data_type> [,  
  <column_name> <data_type> ,  
  .  
  .]  
);
```

[]	Optional
<>	Any

Till ORACLE 21c	A table can have max of 1000 columns only
In ORACLE 23ai	A table can have max of 4096 columns

Data Types in ORACLE SQL:

Data Type tells,

- **How much memory should be allocated**
- **Which type of data should be accepted**

ORACLE SQL provides following data types:

Character related • These are used to hold Strings • String => a group of chars Examples: 'RAJU' , 'INDIA', 'HYD'	Char(n) Varchar2(n) Long CLOB nChar(n) nVarchar2(n) nCLOB
Integer related • These are used to hold integers. • Integer => num without decimal places Examples: 1234, 78, 567, 17	Number(p) Integer Int

Floating point related • These are used to hold float values. • Float => num with decimal places Examples: 20000.00, 5000.80, 67.89	Number(p, s) Float Binary_float Binary_double
Date and time related • These are used to hold date or time values. Examples: 17-FEB-24 6-MAR-25 10.30.15.123456 AM	Date Timestamp [oracle 9i]
Binary related • These are used to maintain multimedia objects like Images, audios, videos, animations, documents, ..	BFILE BLOB

Character Related data types:

All character related data types can accept letters, digits and special chars

Char(n):

- It is used to hold strings.
- It is fixed length data type.
- It is used to hold fixed length chars.
- Extra memory will be filled with spaces.

Varchar2(n):

- It is used to hold strings.
- It is variable length data type.
- It is used to hold variable length chars.
- Extra memory will be removed.

T1		
	F1 CHAR(10)	F2 VARCHAR2(10)
10	RAJU6spaces	RAJU
10	NARESH4spaces	NARESH
		4
		6
Fixed length		Variable length

Examples:

Country_Code CHAR(3)

IND
AUS
WIN

ENAME VARCHAR2(10)

RAJU
NARESH

PAN_CARD_NUMBER CHAR(10)

ABCDE1234F

Mail_id VARCHAR2(30)

Raj_kumar@gmail.com

sai@yahoo.com